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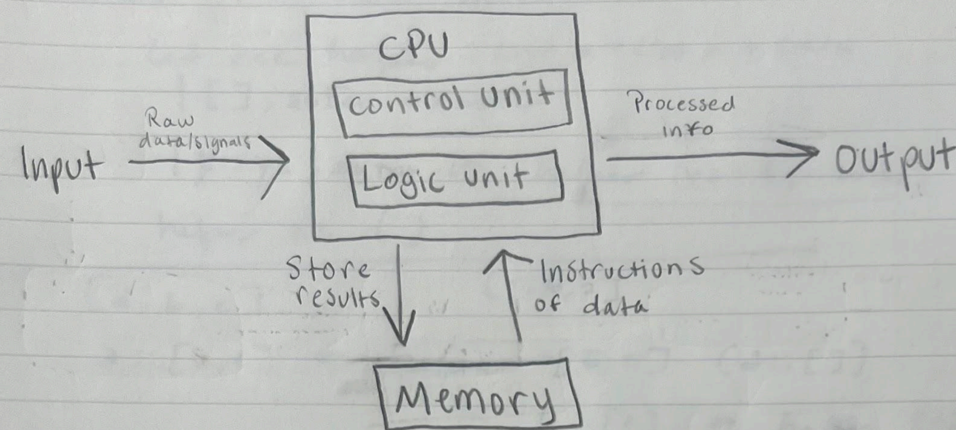
Introduction to compilers and interpreters

1. Draw the Vonn Neumann diagram for computers.



2. Two reasons to study programming languages and compilers:

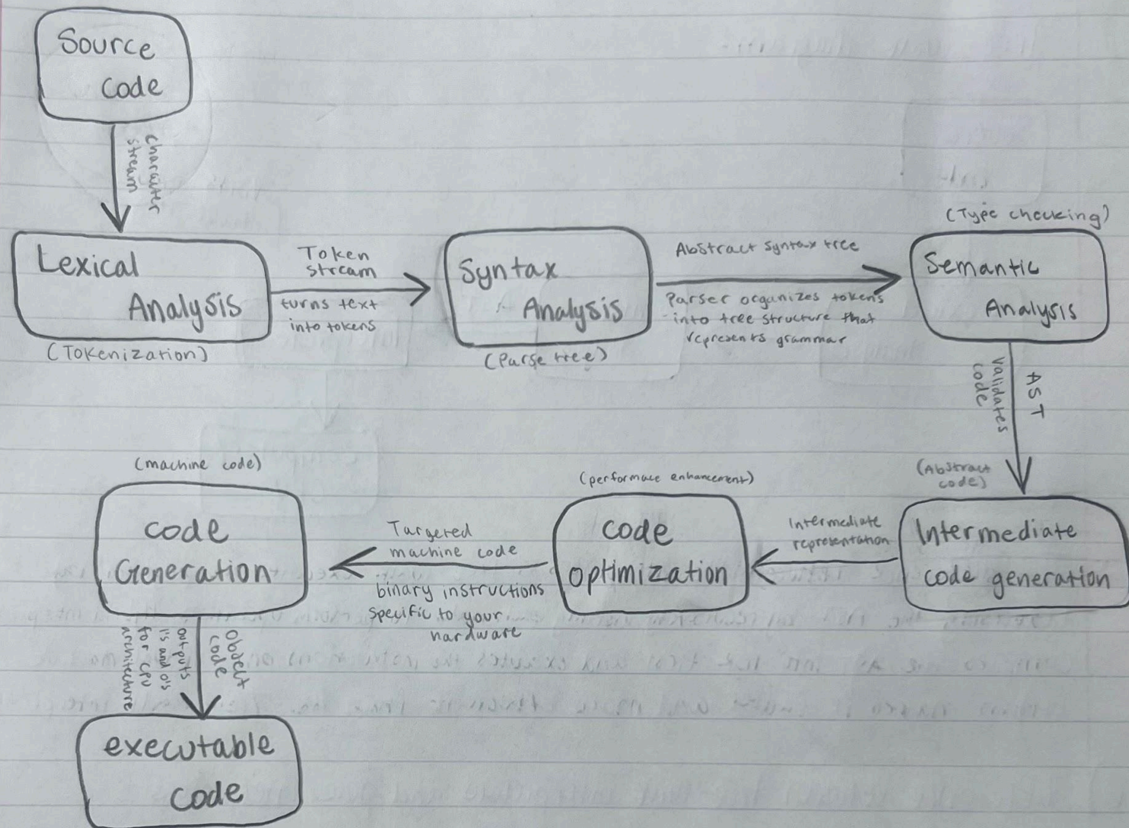
2a) Improved system design and understanding

– Similar to the idea of limited vocabulary is attributed to limited expression of ideas, having a limited knowledge of languages and how they work, limits the systems we can design and understand. By understanding a wider range of programming constructs, we can design reselient complex systems more effeciently.

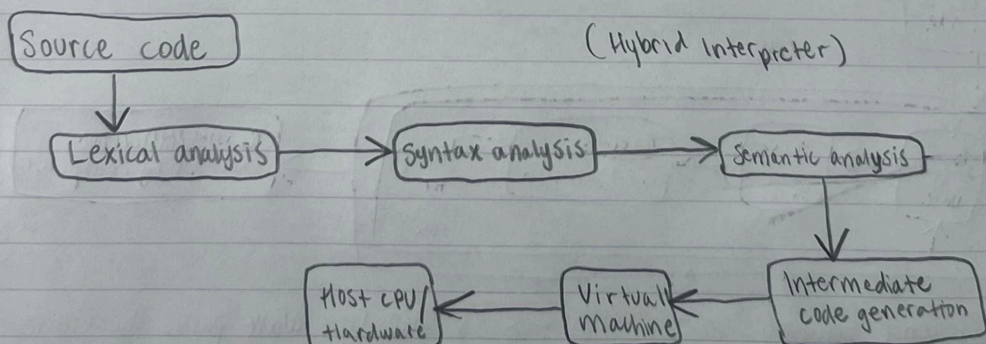
2b) Better understanding of implementation tradeoffs

– Not only is it important to understand what certain languages do and what they're used for, it is also very important to understand why one is better for a job than the other at the lowest level of a computer. Understanding things like how memory is managed, the over head of certain tasks, or even why a certain loop works better for a task saves time, performance, and resources costs.

3) Draw the diagram for a compiler



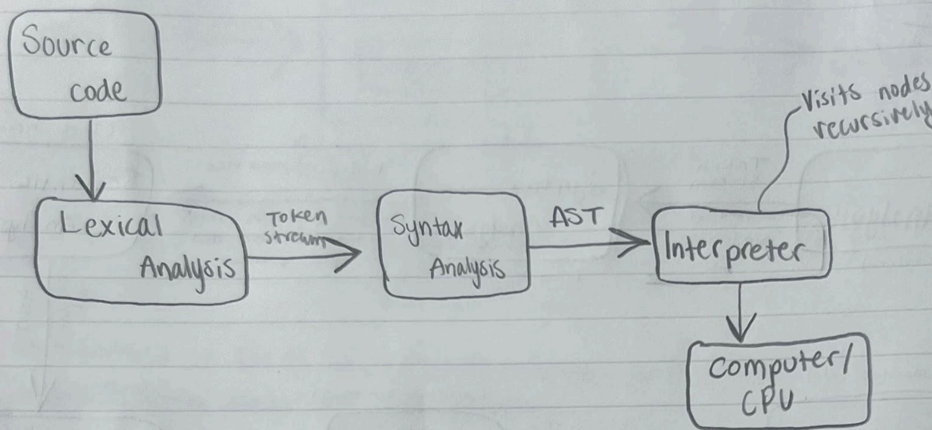
4)



- The main difference between a hybrid interpreter is compilers turn source code into machine code while hybrid interpreters translate source code into IRC which is then interpreted and executed by a virtual machine.

5) difference between hybrid interpreter and Tree-walk interpreter

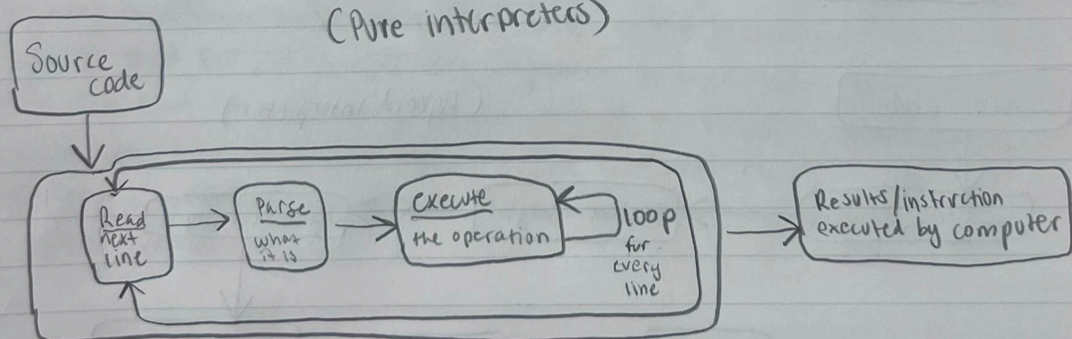
Tree-Walk diagram:



- The difference between the interpreters is tree-walk executes the program by traversing the AST by recursively visiting each node to perform operations. Hybrid interpreters compiles the AST into ICR first and executes the instructions on a virtual machine which makes it faster and more effecient than the Tree walk interpreter.

6) difference between tree walk interpreters and pure interpreters

(Pure interpreters)



- The difference between these interpreters is the tree walk parses the entire program into an AST once before running it. Pure interpreters reads a line, parses it, then executes each line in the program.